(1) Test whether the function $f(x) = \sqrt{x-1}$ is differentiable at $x = 1$.

Solution : $f(x) = \sqrt{x-1}$

$\therefore\ f(1)=0$ and $f(1+h)=\sqrt{h}$

$$f'(1) = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0}\frac{\sqrt{h}-0}{h}$$

$$= \lim_{h\to 0}\frac{1}{\sqrt{h}},$$ which does not exist.

$\therefore\ f'(1)$ does not exist. $\therefore\ f(x)$ is not differentiable at $x=1$.

2) Test whether the function $f(x) = (2x-3)^{\frac{3}{2}}$ is differentiable at $x=\frac{3}{2}$.

Solution : $f(x) = (2x-3)^{3/2}$

using first principle $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

at $a = 3/2$ $f'(3/2) = \lim_{h\to 0} \frac{f(3/2+h)-f(3/2)}{h}$

$f'(3/2) = \lim_{h\to 0} \frac{[2(3/2+h)-3]^{3/2} - [2(3/2)-3]^{3/2}}{h}$

$f'(3/2) = \lim_{h\to 0} \frac{(3+2h-3)^{3/2} - (3-3)^{3/2}}{h}$

$f'(3/2) = \lim_{h\to 0} \frac{(2h)^{3/2}}{h} = \lim_{h\to 0} \frac{2^{3/2} h^{3/2}}{h}$

$\therefore f'(3/2) = \lim_{h\to 0} 2^{3/2} h^{3/2-1} = \lim_{h\to 0} 2^{3/2} h^{1/2}$

$\therefore f'(3/2) = 2^{3/2} (0)^{1/2} = 2^{3/2} (0) = 0$

$\therefore$ limit exists.

$\therefore$ the function $f(x) = (2x-3)^{3/2}$ is differentiable at $x = 3/2$

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3) Show that the function $f(x)$ is continuous but not differentiable at $x = 3$, where

$$f(x)=\begin{cases} x^2+x+3, & for\ x<3 \\ x^2+3x-3, & for\ x\geq 3 \end{cases}$$

Solution :

$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-}(x^2+x+3)$

$\lim_{x\to 3^-} f(x) = (3)^2+3+3 = 15$ — (I)

$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^+}(x^2+3x-3)$

$\lim_{x\to 3^+} f(x) = (3)^2+3\times 3-3 = 15$ — (II)

For $x=3$, $f(x) = x^2+3x-3$

$\therefore f(3) = 9+9-3 = 15$ — (III)

From (I), (II), (III)

$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = f(3)$

$\therefore f(x)$ is Continuous at $x=3$

Now $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

$LF'(3) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h}$

$LF'(3) = \lim_{h\to 0} \frac{(3+h)+(3+h)+3-15}{h}$ [by (III)]

$L.F(3) = \lim_{h\to 0} \frac{9+6h+h^2+h-9}{h}$

$LF'(3) = \lim_{h\to 0} \frac{h^2+7h}{h}$

$LF'(3) = \lim_{h\to 0} \frac{\cancel{h}(h+7)}{\cancel{h}} = 7$ — (IV)

$RF'(3) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h}$

$RF'(3) = \lim_{h\to 0} \frac{(3+h)^2+3(3+h)3-15}{h}$ [by (III)]

$RF'(3) = \lim_{h\to 0} \frac{9+6h+h^2+9+3h-18}{h}$

$RF'(3) = \lim_{h\to 0} \frac{h^2+9h}{h}$

$RF'(3) = \lim_{h\to 0} \frac{h(h+9)}{h} = 9$ — (V)

From (IV) and (V)

$LF'(3) \neq RF'(3)$

$\therefore f(x)$ is not differention at $x=3$

4) Show that the function $f(x)$ is continuous and differentiable at $x = -2$, where $f(x)=\begin{cases} 3x^2-4x+3, & for\ x<-2 \\ x^2-12x-5, & for\ x\geq -2 \end{cases}$

Solution : $\lim_{x\to -2^-} f(x) = \lim_{x\to -2^-}(3x^2-4x+3) = 3(-2)^2-4(-2)+3$

$\therefore \lim_{x\to 2^-} f(x) = 12+8+3 = 23$ — (I)

$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+}(x^2-12x-5) = (-2)^2-12(-2)-5 = 23$ — (II)

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for $x=2$, $f(x)=x^2-12x-5$

$f(-2)=(-2)^2-12(-2)-5=23$ - (III)

From (I), (II), (III)

$\lim_{x\to 2^-} f(x)=\lim_{x\to -2^+} f(x)=f(-2)$

$\therefore f(x)$ is Continuous at $x=-2$

Now $f'(a)=\lim_{x\to 0}\frac{f(a+h)-f(a)}{h}$

$Lf'(-2)=\lim_{h\to 0}\frac{f(-2+h)-f(-2)}{h}$

$Lf'(-2)=\lim_{h\to 0}\frac{3(-2+h)-4(-2+h)+3-23}{h}$ [from (III)]

$Lf'(-2)=\lim_{h\to 0}\frac{3(4-4h+h^2+8-4h+3-2)}{h}$

$Lf'(-2)=\lim_{h\to 0}\frac{12-12h+3h^2-4h-12}{h}$

$Lf'(-2)=\lim_{h\to 0}\frac{3h^2-16h}{h}=\lim_{h\to 0}\frac{h(3h-16)}{h}$

$Lf'(-2)=\lim_{h\to 0}(3h-16)=-16$ - (IV)

$Rf'(-2)=\lim_{h\to 0}\frac{f(-2+h)-f(-2)}{h}$

$Rf'(-2)=\lim_{h\to 0}\frac{(-2+h)^2-12(-2+h)-5-23}{h}$ [by (III)]

$Rf'(-2)=\lim_{h\to 0}\frac{4-4h+h^2+24-12h-28}{h}$

$Rf'(-2)=\lim_{h\to 0}\frac{h^2-16h}{h}=\lim_{h\to 0}\frac{h(h-16)}{h}$

From (IV) and (V) $Lf'(-2)=Rf'(-2)$

$\therefore f(x)$ is diff. at $x=-2$

5) Let $y=\sin^2 x$. Find $\frac{dy}{dx}$ using method of first principle. Hence find $\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{4}}$.

Solution : Let $y=f(x)=\sin^2 x$

By first principle

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}\frac{\sin^2(x+h)-\sin^2 x}{h}$

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}\frac{[\sin(x+h)-\sin x][\sin(x+h)+\sin x]}{h}$

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}\frac{2\cos\left(\frac{x+h+x}{2}\right)\sin\left(\frac{x+h-x}{2}\right)[\sin(x+h)+\sin x]}{h}$

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}\frac{2\cos\left(\frac{2x+h}{2}\right)\sin(h/2)[\sin(x+h)+\sin x]}{h}$

$f'(x)=\frac{dy}{dx}=\lim_{h\to 0}2\cos\left(\frac{2x+h}{2}\right)\frac{\sin(h/2)}{(h/2)}\times\frac{1}{2}\times[\sin(x+h)+\sin x]$

Using $\lim_{h\to 0}\frac{\sin\theta}{\theta}=1$

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$\therefore f'(x) = \frac{dy}{dx} = 2\cos\left(\frac{2x+0}{2}\right) \times 1 \times \frac{1}{2}[\sin(x+0)+\sin x]$

$f'(x) = \frac{dy}{dx} = \cos x \times 2\sin x$

$\therefore \left(\frac{dy}{dx}\right)_{x=\pi/4} = \cos\frac{\pi}{4} \times 2 \times \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} \times 2 \times \frac{1}{\sqrt{2}} = 1$

$\therefore \left(\frac{dy}{dx}\right)_{x=\pi/4} = 1$

6) Let $f(x) = \begin{cases} x^2, & for\ x<2 \\ 4x+2, & for\ x \geq 2 \end{cases}$

Evaluate the left derivative and right derivative of $f(x)$ at $x=2$.

Is $f(x)$ differentiable at $x=2$? Is $f(x)$ continuous at $x=2$?

Solution : $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-}(x^2)$

$\therefore \lim_{x\to 2} f(x) = (2)^2 = 4$ —— (I)

$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+}(4x+2)$

$\lim_{x\to 2^+} f(x) = 4\times 2+2 = 10$ —— (II)

For $x=2$, $f(x) = 4x+2$

$\therefore f(2) = 4\times 2+2 = 10$ —— (III)

from (I), (II), (III)

$\lim_{x\to 2^+} f(x) \neq \lim_{x\to 2^+} f(x) \neq f(2)$

$\therefore$ The given function $f(x)$ is not Continuous at $x=2$

Now $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

$Lf'(2) = \lim_{h\to 0} \frac{f(2+h)-f(2)}{h}$

$Lf'(2) = \lim_{h\to 0} \frac{(2+h)^2-10}{h}$

[by (III)]

$Lf'(2) = \lim_{h\to 0} \frac{4+4h+h^2-10}{h}$

$Lf'(2) = \lim_{h\to 0} \frac{h^2+4h-6}{h}$

Here L.H. derivative does not exist

$\therefore$ The given function $f(x)$ is not differentiable at $x=2$

thus the given function $f(x)$ is neither continuous nor differentiable at $x=2$

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